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# BIOCHEMISTRY



## PAPER 1:

• Enzyme and Membrane Transporters •
Chemistry and Metabolism of Carbohydrates •
Chemistry and Metabolism of Lipids • Chemistry

and Metabolism of Proteins • Nucleotide Metabolism.

• AETCOM

## <u>PAPER 2:</u>

• Molecular Biology
• Nutrition
• Extracellular Matrix
• Biological Oxidation
• Oncogenesis and Immunity.



Referred From VASUDEVAN (10th EDITION)

# CELL & MEMBRANE TRANSPORT

## GROUP-A

1. Discuss how macromolecules are transported across plasma membrane. Indicate the different mode of glucose transport in different organs along with factors responsible for its enhancement. [6+6][2010-S] PAGE 20-26,146-148 2. Classify the membrane proteins. Mention their functions. Explain the ion channel as a mode of transport across the plasma membrane.[3+3][2012-S] PAGE 18-19,21-23

3. Discuss how the fluidity of plasma membrane largely depends on its lipid composition. Describe how the macro molecules are transported across the plasma membrane. Explain the role of ion channel, lipid rafts and caveolae.[6+4+2] [2017-S]

PAGE 18,20-26,21,19

4. Discuss how macromolecules are transported across plasma membrane with schematic diagram wherever applicable[8+4][2018-S] PAGE 20-26

# GROUP-B

1. Describe the biophysical principles involved in the dialysis of blood. Mention its clinical significance. [5+2][2010-S] PAGE 9-10
2. Discuss the role of phospholipid, cholesterol and carbohydrates in the structural and functional aspect of plasma membrane.[2016-S] PAGE 18 3. State different types of transport of molecules across biomembrane. Mention characteristics of Carrier Mediated Transport.Differentiate between primary active transport and secondary active transport.[2+2+3][2019-S] PAGE 20-26 4. Describe the process used to separate cell organelles and list the marker enzymes of cell membrane and sub-cellular organelles. (6+4) [2024]

# GROUP-C[SN]

1. Marker enzyme for subcellular fraction[2010-S] PAGE 13
2. Protein: Lipid is maximum inner mitochondrial membrane[2012-S] PAGE 17 3. Cytoskeleton Structure[2013-S] PAGE 20
4. Receptor mediated endocytosis[2014] PAGE 25-26
5. Osmosis[2014-S] PAGE 8-9
6. Ionophores [2014] PAGE 22-23
7. Secondary active transport [2017-S] PAGE 23
8. Channelopathies.[2025]
9. Segregation and identification of cell organelle.[2025]
10. Hepatic Porphyria.[2025]



# GROUP-D[EQ]

1. Colloids are biologically important having clinical significance.[2013] 2. Some enzymes play important role in identification of cellular organelles.[2015-S] 3. The mechanism of facilitated diffusion can be explained by a ping-pong model[2015-S] 4. Oral Rehydration Solution contains glucose.[2019]
5.Physical techniques are used to isolate subcellular organelles. [2023] 6. G6PDH deficiency leads to haemolytic anaemia.[2025]

7.Haemolysis should be avoided in blood samples collected for potasium estimation.[2025]

# CARBOHYDRATE CHEMISTRY - DIGESTION & ABSORPTION

## GROUP-B

1. Indicate in details the chemical composition of glycosaminoglycans and proteoglycans. Name the carbohydrates present in glycoproteins and glycolipids. [5+2]['13, '15]

2. Describe various form of isomerism exhibited by carbohydrates.[7][2013-S] 3. Describe how monosaccharides and amino acids are absorbed from gut. [2014-S,2016- S]

4. Classify polysaccharides. Indicate the structural and functional aspect of each of them.[2016-S]

5. Describe the various forms of isomerism exhibited by carbohydrates. Name the carbohydrates present in glycoproteins.[5+2][2017]

6. Describe the different types of bond present in heterogeneous polysaccharides with example. [2017-S]

7. Write down the oxidative phase of HMP shunt pathway & its importance. [6+1][2019-S] 8.Explain why people suffer from fasting hypoglycaemia in Von Gierke's disease. Describe how glycogen metabolism differs in skeletal muscles and liver. [4+6] [2022]

## GROUP-C[SN]

1. Blood Group Antigen.['10,'16]
2. Glycosaminoglycans[2012-S]
3. Mutarotation of carbohydrates[2015-S]
4. Glycemic Index['17]
5. Invert Sugar['17]
6. Glucosetransporter['17]
7. Composition And Function of Hyaluronic Acid[2017-S]
8. Discuss isomerism of glucose. [2018-S]
9. Glycosaminology['19

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] 10.Glycosides
[2019-S]

11. RBC Group Antigen[2019-S]

12. Write down the significance of glycosaminoglycans in health.[2023] 13. 4.Mention the structure of glucose transporter in membrane bound state.Give at least three examples of glucose transporters mentioning the tissue they work for.[2025]

14. Renal regulation of maintain pH of blood.[2025]

15.

Glycemic index:Its relevance in diabetes and obesity.[2025]

## GROUP-D[EQ]

1. Sucrose is invert sugar.[2010-S]
2. Glucose and fructose form similar osazone crystals. [2011,2018]
3. Defective lactose digestion may lead to a clinical condition.[2015]
4. Hyperuricemia occurs in Von-Gierke disease.[2018]
5. Thiamine deficiency is detected by measuring transketolase activity in blood [2018] 6. Benedict test is used to identify reducing sugar.[2018-S]

# LIPID CHEMISTRY - DIGESTION & ABSORPTION

## GROUP-A

1. Classify phospholipids with examples. Mention their specific role in maintaining the fluidity of plasma membrane.[10+2][2013, 2010-S 7 marks, 2018 7 marks] 2. Classify phospholipids with examples. Indicates the structure and function of a surfactant. Discuss the role of phospholipids in maintaining the fluidity of plasma membrane.[5+2+5][2015-S]
3. Name the membrane phospholipids. Draw the structure of lecithin. Write the products formed by the action of different types phospholipases on lecithin. State the physiological role of lysophospholipids & fatty acids produced by the breakdown of lecithin.[3+1+4+4][2019-S]

## GROUP-B

1. Tabulate a detailed account of chemical composition of plasma lipoproteins.[7][2010] 2. Describe how lipids are digested and subsequently reabsorbed from the gut[5+2][2010-S] 3. Describe the digestion and absorption of TAG with diagram.[2012-S] 4. Classify the fatty acids in details & indicate their physical properties.[5+2][2010-S, 2017]
5. Indicate the chemical composition and methods of separation of plasma lipoproteins. [2017-S]
6. Name ketones bodies. Outline the steps of synthesis & utilisation of ketone bodies.[1+6][2019-S] 7.Oligomycin and cyanide can inhibit respiratory chain.Give the mechanism of inhibition by both



agents.[2025]
8.Functions of Cyclo-oxygenase and role of Cyclo-oxigenase inhibitors as drug.[2025].

## GROUP-C[SN]

1. Separation & identification of lipid by thin layer chromatography['13] 2.

Omega-3 fatty acids.[2015]
3. Glycosphingolipids[2017]
4. Sialic Acid[2017-S]
5. Plasmalogens [2017-S]
6. Sphingomyelin[2018-S]
7. Eicosanoids[2019]
8. Phospholipase[2019-S]

## GROUP-D[EQ]

1. Arachidonic acid may not be considered as an essential fatty acid.[2010] 2. Lecithin is amphipathic as well as amphoteric in nature.[2014] 3. Acid number helps in the identification of rancidity in fats and oils.[2016] 4. Apolipoprotein is a ligand for cell receptors.[2016-S]
5. Trans fatty acid is injurious to health.[2018-S]
6. Dipalmitoyllecithin acts as surfactant of alveolar fluids[2019]
7. Cholesterol is essential for digestion of lipids[2021]
8. Defects in G protein mediated pathways may lead to diseases like cholera.[2024]

# PROTEIN CHEMISTRY - AMINO ACIDS, HAEMOGLOBIN, PLASMA PROTEIN

# GROUP-A

1. Discuss the four orders of protein structures. Describe the Alpha Helical form of a globular protein. State briefly how the amino acid sequence in a polypeptide chain can be determined.[6+2+4][2010]
2. Discuss briefly how the chemical structures of myoglobin and haemoglobin influence their biological activities. Describe the changes that take place in haemoglobin on oxygenation.[6+6][2010]
3. Describe the peptide bond. What are the different forces that stabilise the protein structure at the different levels of organisation? Give an example to explain the primary structure that determines the functional state of proteins.[4+5+3][2011] 4. Describe how the amino acid composition, N-terminal & C-terminal residues of a protein are determined & identified. Describe the bonds responsible for the four structures of proteins. Briefly indicate how a molecular weight of a protein



is determined.[7+3+2][2013]
5. Describe the salient features of alpha helix and beta pleated sheet structure of proteins. Mention the non-covalent interactions which stabilise protein confirmation. Briefly discuss the role of peripheral & integral proteins in the network of plasma proteins.[4+3+5][2014]
6. Indicate how proteins and peptides are purified prior to its analysis. Describe in detail how the number, kind and sequence of amino acids In a polypeptide chain are

determined.[4+8][2015-S]




7. Compare and explain the oxygen binding curves of haemoglobin and myoglobin. Indicate the conformational changes that occur in haemoglobin on oxygenation. Mention the basic variations in the chemical structures of HbS and HbM as compared to the adult hemoglobin.[6+3+3][2014][2016]
8. Describe the bonds responsible to mention the four orders of protein structure. Describe the physical methods by which the molecular weight of a protein can be determined. Explain how a polypeptide can be synthesised to in the laboratory[6+4+2][2017-S]
9. Describe in detail how the number, kind and sequence of amino acids in a polypeptide chain are determined.[4+4+4]
10. Describe the methods of determination of primary structure of proteins. [2018]
11.Classify protein on the basis of their biological function and give one example of each protein. Compare and contrast the structure of keratin, myoglobin and haemoglobin. Draw O2 dissociation curve of HbA (adult haemoglobin) and HbF (foetal haemoglobin) and explain the difference between them.[3+6+3][2019]
12.Write down different levels of organisation of protein. Write down the steps of haem

degradation.{5+5][2021]

13. A 5 year old child came to paediatric OPD with complaints of swelling around the eyes and both legs and generalised body swelling for last 2 weeks. The mother complained that there was diminished urine output along with passage of frothy urine for same duration. On examination: pedal edema+++ Laboratory examination showed: Serum albumin 2.9 g/dl and total cholesterol 348 mg/dl and total urinary protein 2.8g/day. (urinary protein dipstix=+++) What is your provisional diagnosis? Enumerate five plasma proteins and write down their major functions. What are the acute phase proteins? 5+5+5 [2022]

# GROUP-B

1. Describe the principles of electrophoresis. Illustrate with diagram the electrophoretic separation of the serum proteins indicating the significance of each separated band. Explain the importance of acute phase reactants. [3+2+2][2014] 2. Classify L-amino acids present in the proteins. Explain how amino acids are separated and identified from a mixture of amino acids. [2+5][2015] 3. Explain why haemoglobin is called an allosteric protein. Describe how its allosteric structure helps haemoglobin in cooperative binding of oxygen.[3+4][2015-S]

4. Write down the synthesis of bilirubin. Explain the term direct bilirubin, indirect bilirubin and Van der Berg reagent. Mention the changes that take places in serum bilirubin (direct) and bilirubin(indirect) level in haemolytic and obstructive jaundice.[4+2+1][2019]

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5. Outline the process of synthesis of ammonia in human system. State different routes of disposal of ammonia from human body.[4+3][2019-S]

6. Describe how glycogen metabolism differs in skeletal muscles and liver. b) Write down the steps of heme synthesis. Add a note on various porphyrias.[4+6] [2022]

# GROUP-C[SN]

1. Gamma Glutamyl Cycle[2011-S]

2.

Glycosylated Hemoglobin.[2011,2018]
3. Prions.[2011]
4. Bohr effect[2013-S]
5. Optical isomerism of amino acids[2014-S]
6. Isoelectric pH[2014-S]
7. Selenocystine.[2015]
8. Beta pleated sheet.[2017]
9. Classification of amino acid[2018-S]
10. Discuss secondary structure of protein[2018-S]
11. Peptide bond[2019-S]
12. Transamination reaction.[2024]

13. Immunoglobulin M;Structure and functions.[2025]

# GROUP-D[EQ]

1. 2,3BPG helps in delivery of Oxygen to the tissues.[2011]
2. Glycine solution cannot rotate the plane of plane polarised light.[2012]
3. Patient with Hb-S often suffers from anemia.[2013][2017]
4. Myoglobin does not exhibit Bohr effect.[2015]
5. 2,3 BPG plays an important role in stabelising T structure of hemoglobin.[2015-S]
6. Chaperons play a very significant role in protein folding.[2016, 2018]
7. Hb-A1c provides valuable information for management of diabetes mellitus.[2016]
8. HbF has more affinity towards oxygen than HbA.[2019-S]
9. Haemoglobin is supposed to have all four levels of protein structure.[2021]
10. The oxygen dissociation curve for myoglobin and haemoglobin suit their respective physiologic roles. [2023]
11. Homocysteine is related to atherosclerotic vascular disease and thrombosis [2023]
12. Histidine has an important role in buffering action of proteins.[2024]

## ***ADVANCE TOPICS***

1. Why Phototherapy is used to cure neonatal/ physiological jaundice
2. Beta Thalassemia



# ENZYMES

# GROUP-A

1. Discuss the role of substrate concentration in enzyme catalysed reaction. Indicate

how metas influence this reaction. Discuss how the enzymes are controlled in metabolic pathways. [4+2+6][2010-S]

2. Explain the Michaelis Menten equation and explain the role of substrate concentration on the rate of enzyme catalysed reaction with the help of graphs. Illustrate how Vmax and Km are affected by competitive and non-competitive inhibition of enzymes.“The Km value for glucokinase is much higher than that for hexokinase though both act on glucose” explain the statement.[6+4+2][2017][2013] 3. Define isoenzymes. Discuss the isoenzyme pattern of lactate dehydrogenase and creatine kinase in relation to their role in clinical diagnosis. Explain the principles by which these isoenzymes can be separated and identified in a laboratory.[1+4+4+3][2013-S]

4. Name 5 enzymes whose catalytic activities are altered by covalent phosphorylation-dephosphorylation and indicate their functions. According to the

International Union of Biochemists, enzymes are classified into six major groups. Indicate in which groups the following enzymes belong: i)Adenylate Cyclase, ii)DNA dependent RNA polymerase. iii)Aldolase, iv)Chymotrypsin, v)Reverse Transcriptase, vi)Enolase vii)Acetyl

CoA carboxylase.[5+7][2017][2015] 5. Describe different types of enzyme inhibition. Write the clinical importance of enzyme inhibitors.[8+4] [2018]

6. a) State the class in which the following enzyme do belong: i) Acetyl Carboxylase, ii)Fumarase, iii)Phosphoglucomutase, iv)Aldolase, v)Pepsin, vi)Restriction Endonuclease.

b)Classify the regulatory enzyme. Explain the process of covalent regulation of rate limiting enzyme with suitable example.

c) State at least one pathological condition with rise in activity of the following enzymes in blood: i)SGPT ii)Alkaline Phosphatase, iii)Amylase, iv)RBC Transketolase, v)Creatine Phosphokinase, vi)LDH [3+6+3][2019] 7. A)State at least one pathological condition which increase the activity of following enzyme in blood: Lipase, CPK-MB, RBC glutathione reductase, SGOT B) Explain with the help of enzyme velocity curve how following factors regulate the enzyme activity: Concentration of enzyme, Concentration of substrate, pH, Temperature.
[4+8][2019-S]

8. Classify enzymes according to IUB with examples of each. Differentiate between the lock and key model & induced fit model for enzyme catalysis. What factors affect enzyme activity? Briefly discuss different mechanisms of enzyme inhibition.[6+3+2+4][2021]



# GROUP-B

1. Define isoenzymes. Write the clinical significance of serum isoenzymes in cardiac disorder.[2+5][2012-S]

2. Describe in details how covalent modification and repression/depression mechanism can regulate the enzyme action in vivo.[7][2014-S]

3. Explain the mechanism of allosteric regulation of enzyme activity using PFK as an

example. Mention the other mechanisms by which the enzyme action is regulated.[4+3][2016]

4. Describe the oxido-reductase group of enzymes.

5..Creatine kinase (CK)has relevance in diagnosis of more than one condition.In light of the statement answer the following:-

a.Name the isoenzymes of Creatine kinase.

b.Mention the method by which the iso enzymes can be separated. c.Mention at least three conditions where isoenzymes as well as total CK is raised. [2025]

# GROUP-C[SN]

1. Km of enzyme [2012-S]
2. Coenzyme [2012-S]
3. Ribozyme [2014-S][2016-S][2018]
4. Non-functionalenzyme [2016]
5. Michaelis-Menten Equation [2016-S]
6. Isoenzyme [2018-S][2018]

# GROUP-D[EQ]

1. The mode of action of metallo-enzymes and metal activated enzymes are different.[2013] 2. Isoenzymes of Alkaline Phosphatase are of diagnostic significance.[2014] 3. Non-functional plasma enzymes are important only for clinical purposes.[2014] 4. Methotrexate is a competitive enzyme inhibitor.[2014-S]

5. Apolipoproteins are enzyme cofactors.[2014-S]

6. In competitive inhibition, larger amount of substrate can overcome the effect of inhibition.[2014-S]

7. Isoenzyme assay is helpful in the diagnosis of MI.[2015]

8. Metalloenzymes and metal activated enzymes are not similar. [2016-S] 9. Allopurinol is called suicide inhibitor.[2018-S]

10.Coenzymes act as co-substrate in the enzyme catalysed reaction.[2019] 11. The mode of action of metalloenzymes and metal activated enzymes are different.[2022].

12.Aspartate transcarbamoylase is an allosteric enzyme.[2023]

## ***ADVANCE TOPICS***

1.Between competitive and noncompetitive inhibition



2.Urinal used in treatment of gout is an example of suicide inhibition 3.illuminate different markers used for the diagnosis of myocardial infraction

4.Why ALP level increases in obstructive jaundice

5.Describe the mechanism involved in regulation of enzyme activity

# CARBOHYDRATE METABOLISM GROUP-A

1. Describe in a flow diagram the metabolic pathways of glycogen formation and degradation in the body. Describe in a separate chart show cyclic AMP regulates this process by enzyme modification. [4+8][2013]

2. In a flow diagram describe the metabolic steps of glycogenesis and glycogenolysis in muscle and show how cAMP integrates their regulation.[6+6][2010,2018-S]

3. Describe the metabolic steps of citric acid cycle in a flow diagram indicating the enzymes and coenzymes involved and highlighting the steps where the energy is produced. Mention the steps in the cycle which are irreversible in nature. Indicate how propionate is converted to

one of the intermediates of this
cycle.[8+2+2][2013]

4. In a flow diagram, indicate the metabolic steps by which propionate can be converted to glucose and show how key enzymes of gluconeogenesis are controlled.[6+6][2010]

5. Describe in detail how blood sugar level is maintained at constant level in post absorptive state. Use a flow diagram.[2014-S]

6. Discuss briefly how pyruvate is converted to acetyl coA. Mention how the pyruvate dehydrogenase complex is controlled.[6+6][2015-S]

7. On complete oxidation, glucose leads to production of carbon dioxide and water. Mention those metabolic steps where carbon dioxides are evolved. Give a detailed account of enzymes, co-enzymes and control mechanisms involved in these steps.Mention three examples of metabolic reactions where carbon dioxide is utilised in this process.[6+3+3][2016]

8. Describe in detail how glucogenic amino acids are converted into glucose in the body by TCA cycle and reverse pathway of glycolysis. [2016-S]

9. Describe the role of insulin and glucagon on gluconeogenesis process in fed and starved state. [2018-S]

10. Describe differences between the anaerobic and aerobic pathways of glycolysis. Explain how RBCs can continue glycolysis without any synthesis of ATP' Explain how metabolic pathway of galactose and fructose are linked to glycolytic pathways.[6+3+3+3][2021]

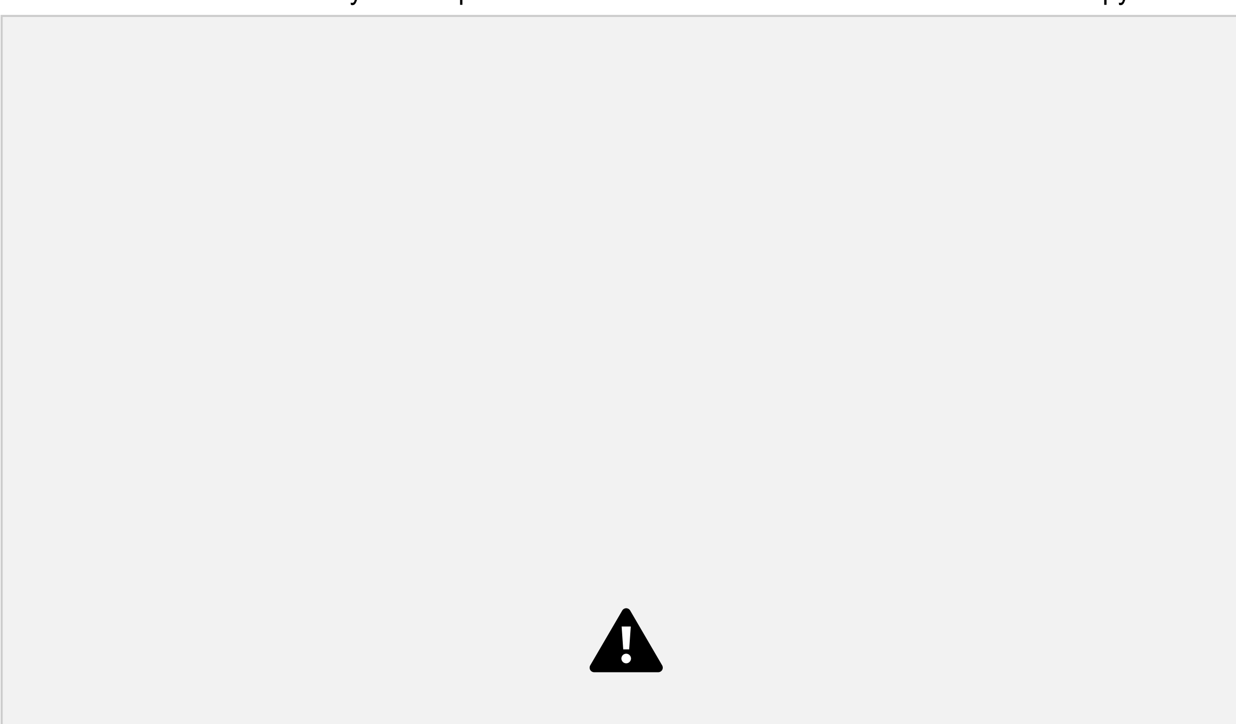
11. A 23-year-old male was brought to the emergency in semiconscious state. His mother gave history of diagnosis of malaria confirmed by identification of parasites in blood smear and subsequent treatment with Primaquine by his family physician. He was also passing dark coloured urine. On examination, he had fever, extreme pallor, severe jaundice, tachycardia and low BP. His sclera was yellow and his spleen was enlarged 1) What is your provisional diagnosis and why? ii) Explain the biochemical basis of the findings in this patient. iti) Outline



the metabolic pathway that is defective in this case. iv) Mention two important functions of this pathway v) Add a note on glutathione and its role in the body.[2+4+4+2+3] [2023]

# <u>**GROUP-B**</u>

1. Describe multi-enzyme complex and various reactions involved in the oxidation of pyruvic



acid to acetyl-CoA.[7][2011]

2. Describe the process of formation of glucose from lactate indicating the regulatory steps. [4+3][2012-S, 2018]

3. Describe in detail how pyruvate is converted to Acetyl-CoA in the body.[7][2013]. 4. Give a brief account of glycogen storage disease.[7][2014].

5. Describe the metabolic steps in hexose monophosphate pathway [7][2015-S] 6. Describe the hormonal control of glycogenesis and glycogenolysis in a flow diagram. [2016-S]

7. Explain with a flow diagram how glycolysis and gluconeogenesis in the liver are controlled by fructose 2,6 bis phosphate & the bifunctional enzyme 6-phosphofructo-2-kinase.[7][2017]

8. With flow diagrams, indicates the normal and abnormal metabolism of fructose and galactose the body.[7][2017-S]

9. Write down the non-oxidative process of HMP Shunt Pathway and its importance. [6+1][2019]

# GROUP-C[SN]

1. Rapoport Luebering cycle['11,'17]
2. Essential Pentosuria ['10,'16]
3. Pyruvate Dehydrogenase['19]

# GROUP-D[EQ]

1. Long chain fatty acids cannot be converted to glucose in human body though the reverse is possible.['10]

2. Impairment of pentose phosphate pathway(PPP) leads to erythrocytic hemolysis.[2011] 3. Phosphofructokinase-I is known as pacemaker of glycolysis.[2012] 4. Von-Gierke's disease is associated with hyperuricemia.[2012][2017] 5. G6PDdeficiencydevelopsanemia.[2012-S][2017-S]

6. Fatcanbesynthesizedfromglucosebutglucosecan'tbesynthesized from fat.[2016] 7. Galactosemic patients are often associated with congenital cataract.[2016] 8. Uncontrolled DM causes cataract. [2018-S]

## ***ADVANCE TOPICS***

1. Give detailed informations about homo Polysaccharides with examples .how can we differentiate between reducing and non reducing sugar

2. Give an account of difference between hexokinase and Glucokinase. Why for estimation of plasma glucose we use fluoride bulb? Add note about regulation of



Glycolysis.

3.Write down the inhibitors, regulation & Amphibolic role of Kreb cycle 4.Name any 4 non-carbohydrate precursors from where formation of glucose ie. gluconeogenesis can be done? Synthesis of Glucose from Propionyl CoA~steps

# LIPID METABOLISM

## GROUP-A

1. Give an account of fatty acid synthase complex. Describe the metabolic pathway for de-novo synthesis of palmitate in the body.[3+9][2014]
2. Describe the metabolic steps of biosynthesis of cholesterol.Discuss the control metabolism associated with HMG CoA reductase. Explain reverse cholesterol transport.[8+2+2][2017]
3. Describe how palmitic acid is oxidised in the body completely and calculate its net gain in energy. Explain how the complete oxidation of oleic acid different from that of palmitic acid.[8+2+2] [2017-S]
4. Write the metabolism of VLDL. Explain the reverse-cholesterol transport. [8+4][2018]
5. Write down the steps of beta-oxidation of fatty acid. Why defective beta oxidation may lead to hypoglycemia. [6+4][2021]
6. Define Lipids. Classify them with examples. Describe the structure and function of

Lipoproteins. Explain the VLDL metabolism in our body. Add a note on dyslipidemia. 1+4+4+4+2 [2022]

7.A 8 year old boy has serum LDL 230mg/dl, HDL 35 mg/dl, VLDL 25 mg/dl, Triglycerides 126 mg/dl. His brother and father had isolated increased LDL cholesterol i) What is your provisional diagnosis? ii) Discuss the Fredrickson classification of hyperlipoproteinemia iii) Mention in a flow diagram, the cholesterol synthesis up to mevalonate iv)Name a lipid lowering agent with its mechanism of action.[2+5+5+3] [2023]

8. A 60 year old person old person is suffering from diabetes for last 10 years. a.List the biochemical investigations you should advise him as a part of regular check-up.
b.Write down the biochemical reference range of those parameters.
c.Explain the alternation of lipid metabolism in diabetes.[2025]

## GROUP-B

1. Explain peroxisomal beta oxidation and its importance.[7][2012-S] 2. Give the exact chemical composition of very low density lipoprotein. Explain their formation and fate inside the body.[2+5][2013]

3. Discuss the role of carnitine in the oxidation of long chain fatty acids. Indicate what happens in their deficiency.[5+2][2013-S]

4. Discuss the process of lipogenesis and lipolysis which take place in adipose tissue indicating the role of insulin and glucagon in this process.[7][2013-S]

5. Describe how ketone bodies are formed & subsequently degraded in the body.[3+4][2015]

6. Describe with the help of a diagram digestion of a triglyceride and its absorption from intestine, with special reference to the role of bile in the

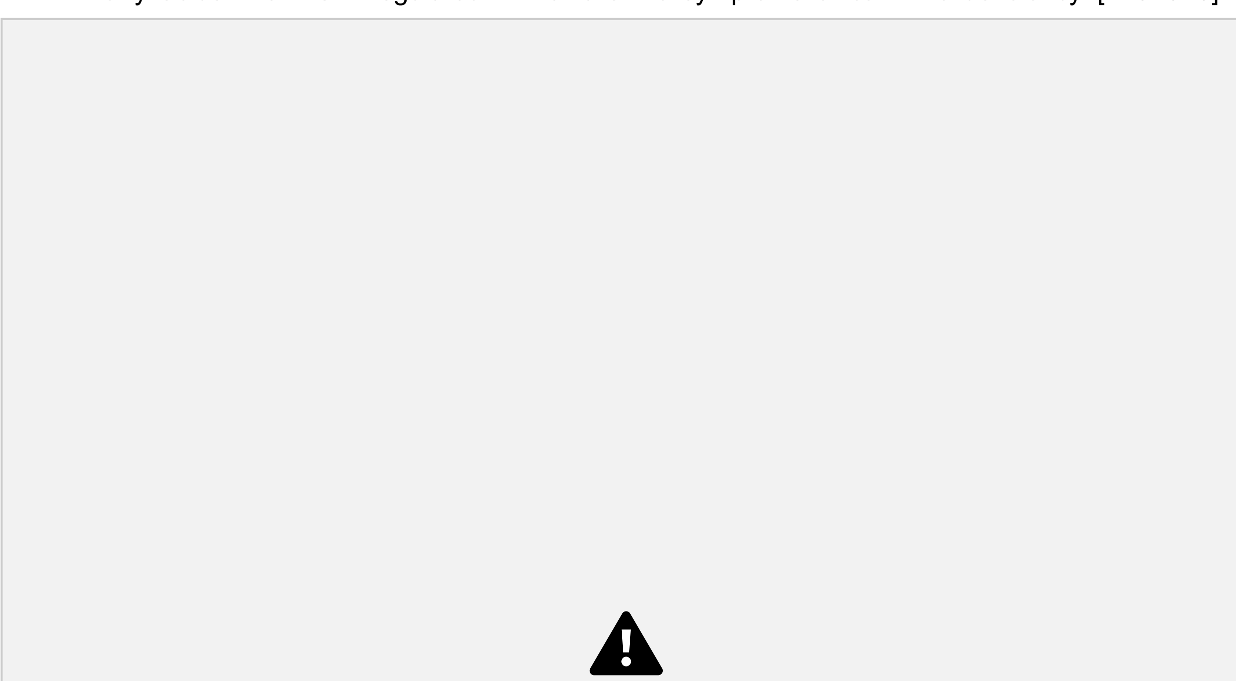


process.[2019]

7. Write down the β-oxidation of palmitic acid and mention problem that arise out of medium chain Acyl CoA Dehydrogenase deficiency.[6+1][2019]

8.What are ketone bodies? How are they formed in the body? Describe the role of ketone bodies in starvation and in severe uncontrolled Diabetes Mellitus.[2+4+4] [2022]

9.Write down the chemical name of carnitine. Describe the role of carnitine in Beta oxidation of fatty acids. How is it regulated? What are the symptoms of carnitine deficiency?[1+3+3+3]



[2023]
10. Name the rate limiting enzyme of fatty acid synthesis. Describe the regulation of it. Discuss how this enzyme's activity influences beta-oxidation of 11. fatty acid. Name the cell organelle where the fatty acids can be desaturated and elongated. (1+5+3+1)[2024]

## GROUP-C[SN]

1. Control of HMG-CoA reductase.[2013]
2. LCAT[2014-S]
3. Role of carnitine in fatty acid metabolism.[2015]
4. Fatty acid synthase complex.[2017]
5. Fattyliver[2017-S]

## GROUP-D[EQ]

1. Defective beta oxidation of fatty acid causes hypoglycemia in infants at night. [2011-S] 2. Statin group of drugs(atorvastatin) act as cholesterol lowering agent.[2012-S, 2019-S] 3. Ketone bodies are degraded in the extrahepatic tissues only. [2013] 4. Both uncontrolled diabetes mellitus and prolonged fasting produce ketosis but its magnitude is less in the case of prolonged fasting. [2014]
5. Ketosis cannot lead to increased cholesterol synthesis. [2014-S]
6. HDL is involved in reverse cholesterol transport. [2015, 2019]
7. Lipoprotein lipase deficiency may lead to hyperglyceridemia. [2015] 8. Consumption of alcohol leads to fatty liver.[2015-S, 2016-S]
9. Impaired beta oxidation of fatty acid may lead to hypoglycemia. [2015-S] 10. Citrate plays an important role in fatty acid synthesis.[2017, 2018-S] 11. Increase intake of fructose leads to formation of more VLDL.[2017-S] 12. Ketone bodies are not waste material. [2018-S]
13. HDL C is good cholesterol.[2018-S]
14.Brown adipose tissue promotes thermogenesis.[2022]
15.Sudden infant death syndrome is seen in defects of beta oxidation.[2022]

## *ADVANCE TOPICS*

1.Classification of fatty acids
2.How fatty acid synthesis takes place
3.Disorders related to lipoprotein metabolism
4.Fatty liver



# PROTEIN METABOLISM

## GROUP-A

1. Name the aromatic amino acids. Outline the catabolic pathway of phenylalanine mentioning the disorders relating to that.Describe how catabolism of haem produces bilirubin. Indicate in details the process of uptake, conjugation and secretion involved in

transfer of bilirubin from blood to bile.[6+6][2014]
2. Describe the formation and fate of catecholamines[7][2015-S]
3. Discuss an essay on biosynthesis of urea. How it is regulated. Add a note on the clinical significance of uremia. [2018-S]
4. Describe the reaction of transamination and oxidative deamination of amino acids in the body. What are the effects of hyperammonemia?[3+3+4][2021] 5. A middle aged woman rushed to emergency with pain abdomen and nausea. On asking patient confirmed dark urine and grey stool. After managing the emergency situation blood samples were tested for liver function. Results of the same as follows total bilirubin-6.2mg/dl, Direct Bilirubin-5.8mg/dl, Total protein 7.2

gm/dl, Albumin-3.8 gm/dl, AST-26 U/L, ALT -32 U/L, Alkaline phosphatase-387 U/L, GGT-35 U/L. i) As an attending medical officer what would be your provisional diagnosis?
ii) What all other tests would you suggest to justify your diagnosis? iii) How would you exclude acute pancreatitis or alcoholic?iv) What are the different causes of Conjugated hyperbilirubinemia?(3+4+4+4)[2024]

6. A patient with chronic renal failure is admitted in comatosed state in hospital. On clinical examination features of encephalopathy are seen. Blood investigation shows creatinine level of 6.5 mg/dl and urea level of 425 mg/dl..
i) Describe the metabolic derangement responsible for this condition. ii) Explain the biochemical basis of ammonia toxicity in the brain.
iii) Describe how the alpha amino group in most of the amino acids are converted into ammonia. 4+7+4[2024]

# GROUP-B

1. Describe how catecholamines are synthesised and degraded inside the human body.[3+4][2010]
2. Write the synthesis, transport and degradation of catecholamines.[7][2011] 3. Describetheprocessoftransaminationandoxidativedeaminationinthe body.[4+3][2014] 4. Explain the role of glutamic acid in removal of ammonia from amino acid. Why ammonia is toxic to the central nervous system. [2018]

5.Mechanism of action of Methotrexate and Ethanol in Methanol poisoning.[2025]

# GROUP-C[SN]

1. Maple Syrup Urine Disease.[2010]
2. Polyamines.[2010]



3. S-AdenosylMethionine.[2013]
4. Acute intermittent porphyria.[2014]
5. Conjugated hyperbilirubinemia[2015-S]
6. Phenylketonuria [2016-S, 2018-S, 2019-S]
7. Hyperbilirubinemia[2016-S,2018]
8. Oxidative & non-oxidative deamination. [2017-S]
9. Alkaptonuria [2019]
10. Transmethylation [2019-S]

11. Oxygen dissociation curve.[2022]

## GROUP-D[EQ]

1. Urine turns black on standing in Alkaptonuria.[2010]
2. Phototherapy (exposure to blue light) helps in treatment of neonatal Physiological jaundice.[2011][2017-S]
3. Ammonia is toxic to Central Nervous System.[2012, 2016-S]
4. Patient with carcinoid syndrome may exhibit pellagra.[2013]
5. Alkaptonuria is often associated with generalised pigmentation of Connective tissue (Ochronosis).[2013]
6. Drugs may precipitate attacks of porphyria in some patients. [2017-S] 7. Hartnup disease gives rise to pellagra like syndrome.[2019]
8. Haemoglobin is supposed to have all four levels of protein structure.[2021] 9.

Allopurinol is used to threat gout.[2025]

10. Misfolding of protein causes disease.[2025]

11. Hypoalbuminemia leads to peripheral oedema.[2025]

## ***ADVANCE TOPICS***

1.Functional classification of proteins
2.Give an account of biologically important peptides
3.Describe the specialised products given by phenyl Alanine, tyrosine 4.Urea cycle disorders
5.Disorders related Tryptophan metabolism in hartnup disease carcinoids syndrome,Xanthurenate in urine.

# BIOLOGICAL OXIDATION

## GROUP-A

1. Define redox potential. Explain its significance in ETC with a schematic diagram. Write the role of inhibitors of ETC.[3+6+3][2012-S]
2. Describe all the complexes with their components of the respiratory chain in mitochondria with the probable sites of ATP synthesis. Indicate the names of different inhibitors with their sites of inhibition. Name some uncouplers associated with respiratory chain and indicate their significance.[8+2+2][2014-S,



2018]
3. Describe the organisation of the electron transport chain mentioning the components of all four complexes and the reaction catalysed by them. Explain the mechanism of ATP synthesis in the mitochondria by ATP synthase. [2016-S]

## GROUP-B

1. Describe the chemiosmotic coupling hypothesis of oxidative phosphorylation.[7][2010] 2. Describe the mitochondrial electron transport chain. How the inhibitors of ETC differ from uncouplers of oxidative
phosphorylation?[5+2][2011]
3. Describe the operation and significance of glycerophosphate shuttle and malate shuttle.[3+4][2014]
4. What is oxidative phosphorylation? Differentiate it from substrate level phosphorylation. Illustrate with a diagram how ATP is synthesised in mitochondria?[2+2+3][2014]
5. Give an account of inhibitors and uncouplers of mitochondrial respiratory chain.[5+2][2015-S]
6. Name the components of the electron transport chain with the help of a diagram. Explain how the electron flows from the NADH+H+ to the molecular oxygen through the different components of ETC. [2+5][2019- S] 7. Describe the mitochondrial electron trans port chain with diagram. Name the inhibitors of different complex of electron transport

chain.[6+4] [2023] 8. Define xenobiotic. Describe the different phases of xenobiotic metabolism with proper examples. 2+8

## GROUP-C[SN]

1. Malate Shuttle[2010-S]
2. Uncoupler of oxidative phosphorylation.[2019]

## GROUP-D[EQ]

1. Certain drugs are inhibitors of respiratory chain in the mitochondria[2010-S] 2. Brown adipose tissue promotes thermogenesis.[2011,'10]
3. Ubiquitin and cytochrome C have special role in respiratory chain reactions as they are mobile carrier of reducing equivalent.[2013-S]
4. G6PD is responsible for erythrocyte membrane rigidity.[2014]
5. Brown fat is responsible for non-shivering genesis in newborn. [2016-S] 6. F0F1 ATPase give rise to ATP synthesis in intact mitochondria.[2019-S] 7. TCA is known as amphibolic pathway.[2021]
8. 2,4 dinitrophenol functions as an uncoupler of the respiratory chain.[2022] 9. The chemiosmotic theory explains the mechanism of oxidative phosphorylation.[2023] 10. Cytochrome C has important role in apoptosis.[2024]

### CLINICAL & APPLIED BIOCHEMISTRY

# ACID-BASE BALANCE



## GROUP-A

1. Describe in detail how pH of the blood is regulated by lung and kidney. Give an account of metabolic acidosis and anion gap.[4+4+2+2][2014-S, 2018-S]

## GROUP-B

1. Describe the renal mechanism for regulation of acid base balance. 2.
What is the biomedical importance of anion gap?[4+3][2014]

3. Name the blood buffers.Explain the role of blood buffers in the maintenance of normal pH of blood.[2019]

4. Write down the Henderson Hasselbalch Equation. Explain the role of kidney in the maintenance of acid base balance in our body.[1+6][2019- S] 5.A 50 year old man was admitted to hospital with complaint of persistent vomiting On examination, he was found dehydrated and the respiration was shallow. He gave past H/O dyspepsia. The result of the laboratory investigations are as follows: Parameter Blood pH Plasma HCO 7.72 Obtained Value 45mmol/L Na^{+} 60mmHg 140 E\overline{g}/L
K Urine 2.5 mEq/L Acidic 1) Interpret the report and give a probable diagnosis based on acid base disorder Explain the compensatory phenomenon going on in this state. 1) iii) Explain briefly the cause of hypokalemia with excretion of acidic urine in this patient.[5+2+3] [2023]

## GROUP-C[SN]

1. Respiratory Acidosis [2013]
2. Henderson–Hasselbach equation[2013-S]
3. Alkali reserve[2018]
4. Bicarbonate buffer system[2021]

## GROUP-D[EQ]

1. Histidine plays major role in buffering.[2015-S]
2. Cellular exchange of ion maintains hydrogen ion homeostasis. [2017-S]

# FUNCTIONAL TEST

## GROUP-B

1. Describe the principle and outline of methodology of ELISA. Indicate the significance of the test. [7][2014-S]
2. Describe the methods of determining the chemical structure of any unknown biomolecule.[7][2017]
3. Describe the principles of different forms chromatography and indicate their role in clinical diagnosis. [2017-S]
4. Enumerate the difference between autosomal dominant and recessive



disorders giving examples. [2017-S]

## GROUP-C[SN]

1. Oral GTT.[2011-S]
2. Renal Function Test[2012-S]
3. Electrophoresis.[2016,2018-S]
4. Ion Exchange Chromatography.[2016-S]

5.

Enzyme assay by coupling to a dehydrogenase[2016-S]
6. Abnormalities Of Thyroid Function.[2018-S]
7. Biochemicalfunctionofascorbicacuid.[2018-S]
8. Thin Layer Chromatography.[2019-S]

## GROUP-D[EQ]

1. Renal clearance study is a nearly predictor of impending renal failure.[2014, 2018] 2. Levels of hepatic enzymes can differentiate b/w hemolytic, hepatocellular and obstructive jaundice. [2013]
3. Glycosuria can happen even with absence of hyperglycemia. [2014-S] 4. Urinary urobilinogen is increased in hemolytic jaundice. [2018] 5. Phototherapy is better alternative than Phenobarbital in treatment of Crigler Najjar syndrome type I. [2021]
6. Serum creatinine is more specific than plasma urea for assessing kidney function.[2022]
7. Justify the clinical significance of creatinine clearance rate in renal function.[2024]

# FREE RADICALS, ANTIOXIDANTS & XENOBIOTICS

## GROUP-B

1. Name the dietary antioxidants. Discuss how they protect from oxidative injury?[5+2][2011-S]
2. Describe the role of glutathione, glucuronic acid and glutamine in the process of detoxification. [7][2014-S]
3. Explain the different phases of metabolism of xenobiotics. [7][2016-S]



4. State the phases of biotransformation of xenobiotics. Enumerate four reactions of Phase II giving example of each. Add a note on cytochrome P450 enzyme system. [2+4+4] [2023]
5. Name two vitamins which have role as antioxidants. Briefly describe the sources, their mode of action as antioxidants and deficiency manifestations [2+1+5+2] [2022]

## GROUP-C[SN]

1. Antioxidant Vitamins.[2012-S]
2. Superoxide Dismutase. [2013]
3. Role of Cyt-P450 in hydroxylation reaction.[2013]
4. Lipid Peroxidation Reaction[2013-S][2016-S]
5. Biochemical functions of peroxisomes.[2014]
6. Antioxidant Enzymes.[2014]
7. Free Radical Induced Damage.[2015-S]
8. Reactive Oxygen Species[2019]
9. Cyt-P450[2014-S,2019]
10. Phase II reaction of Xenobiotics.[2019-S]
11. Anti oxidant vitamins and their interdependence.[2024]
12. Define xenobiotics.Explain the role of Cytochrome P450 in Xenobiotic mechanism.[2025]

## GROUP-D[EQ]

1. Some Enzymes Take Part In Scavenging Of Free Radicals.[2010-S] 2. Free radicals may have some beneficial role also[2015-S]
3. Glutathione plays a vital role in detoxification. [2016-S]
4. Hydroxylation reactions often require the presence of ascorbic acid. [2017-S] 5. Macrophages shows beneficial effects by generating free radicals.[2021] 6. Glutathione is an important mediator for detoxification of toxic materials in humans. [2022]

## ***ADVANCE TOPICS***

1.Sources of free radical
2.Disorders related to reactive oxygen species (ROS)
3.Name any three chain breaking antioxidant

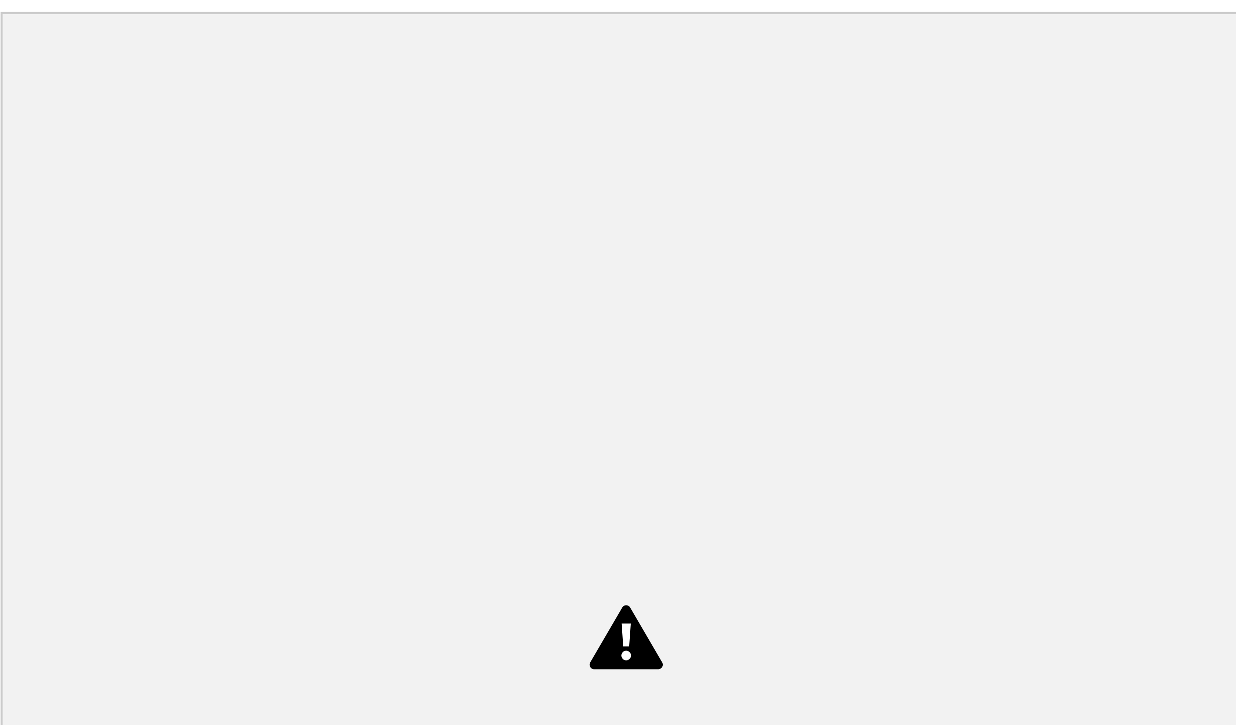
# VITAMIN, MINERALS & NUTRITION

## GROUP-A

1. Explain how iron is absorbed, transported and stored in the body. [12][2012-S][2014-S] 2. What is a balanced diet? What is the nutritional importance of dietary proteins? Discuss the protein energy malnutrition with special reference to kwashiorkor disease.[3+6+6][2021]
3. Define and classify minerals. What are the biochemical functions of calcium in



the body? Write a note on: Gow the homeostasis of plasma calcium level is achieved? [3+6+6] [2023]
4. A 60 year old man reports to the medicine OPD with complaints of extreme weakness and fatigue as well as numbness and heaviness in lower limbs for past 6 months. On taking history it was found that he is a strict vegetarian. On examination of the patient he looks pale and neurological examination showed sensory and motor loss of both the legs.
Findings of the laboratory investigations are: Peripheral smear showing large sized



RBC.Urine showed high levels of methyl malonic acid and homocysteine. 1) Identify the nutritional deficiency and the probable diagnosis.
ii) Correlate the patient symptoms and the signs on examination with the lab reports. iii) Enumerate other lab investigations that can be ordered to reach to a definite diagnosis. iv) Analyse the lab findings and give biochemical explanation of the same. v) Plan a treatment for this patient.(2+3+2+5+3)[2024]

5.A 15 year old high school boy had difficulty in performing his work in dim light.Dietary history revealed that he thrives on fast food like noodles,cakes and biscuits.Based on the history,
a.What is the probable diagnosis?
b.Name the biochemical compound which may be deficient in the above condition.
c.What is the recommened daily allowance of it for children and adults? d.Mention the different functions of this compound.
e.Explain the aforementioneclinical condition.Describe the cycle associated with the compound with diagram. [2025]

## GROUP-B

1. Describe how one carbon compound is utilised in the body mentioning the names of the donors and acceptors of those compounds. [2014-S]
2. Indicate the factors which modified absorption of calcium from the gut. Discuss in detail how calcium metabolism is controlled by calcitriol and parathyroid hormone. Enumerate the biochemical role of intracellular calcium.[3+4] [2016-S] 3. Diagrammatically discuss the absorption transport and storage of Iron. Enlist the inn containing proteins, justify the role of cytochrome in electron transport chain.[6+4] [2022]

4.Effect of folic acid Deficiency on pregnant mother and developing foetus.[2025]

## GROUP-C[SN]

1. Wald's Visual Cycle [2014-S]
2. Folic Acid As Coenzyme [2016-S]
3. Folate Trap [2018]
4. Nitric oxide has various roles. [2018-S]
5. Biochemical Function Of Ascorbic Acid [2018-S]
6. Protein energy malnutrition[2022]
7. Name the proteins related to iron absorption and transport mentioning function of each other.[2025]



## GROUP-D[EQ]

1. Folic acid cures megaloblastic anemia.[2011-S]
2. Vitamin K takes part in blood coagulation [2014-S]
3. Transamination reaction cannot take place without pyridoxine.[2017-S] 4. Vitamin K deficiency is responsible for hemorrhagic disease of newborn.[2019] [2023] 5. Normal function of the kidney is essential for the synthesis of active Vitamin D3. [2019- S]

6.

Deficiency of ascorbic acid causes fragility of blood vessels.[2021]
7. Vitamin B12 should be given along with folic acid to treat folic acid deficient anemia.[2022]
8. Wilson's disease is a disorder of copper metabolism.[2022]
9. Following vegan diet strictly may lead to vitamin B12 deficiency

## ***ADVANCE TOPICS***

1. Why vitamin d is also known as hormone?
2. Give detail informations of vitamin d about sources, synthesis, biochemical function
Vit K cycle
3. One carbon Metabolism
4. Write about RDA ,sources functions, deficiency manifestation of COPPER.

# NUCLEOTIDE CHEMISTRY

# GROUP-A

1. Describe the chemical structure of DNA in detail. Explain the chemical nature of RNA which differs from that of DNA. Describe the structural difference between mRNA & tRNA.[7+3+2][2013-S]
2. Describe the Watson Crick model of DNA structures enumerating its salient features. Explain the role of Histone Protein in the organisation of DNA. How is denaturation of DNA used to analyse its structure? [2016-S]

# GROUP-B

1. Indicate the functional aspect of all varieties of RNA. [2016-S]
2. Describe Watson crick model of DNA structure. Draw and label structure of tRNA[4+3] [2018-S]

# GROUP-C[SN]

1. t-RNA[2011]
2. Synthetic Nucleotide[2013-S]
3. Z DNA[2014-S]
4. Major & minor bases of nucleotides.[2015-S]
5. Small RNA[2015-S]
6. Bonds in polynucleotides.[2017]
7. snRNA, mi RNA, siRNA [2017-S]
8. Melting of DNA [2018]

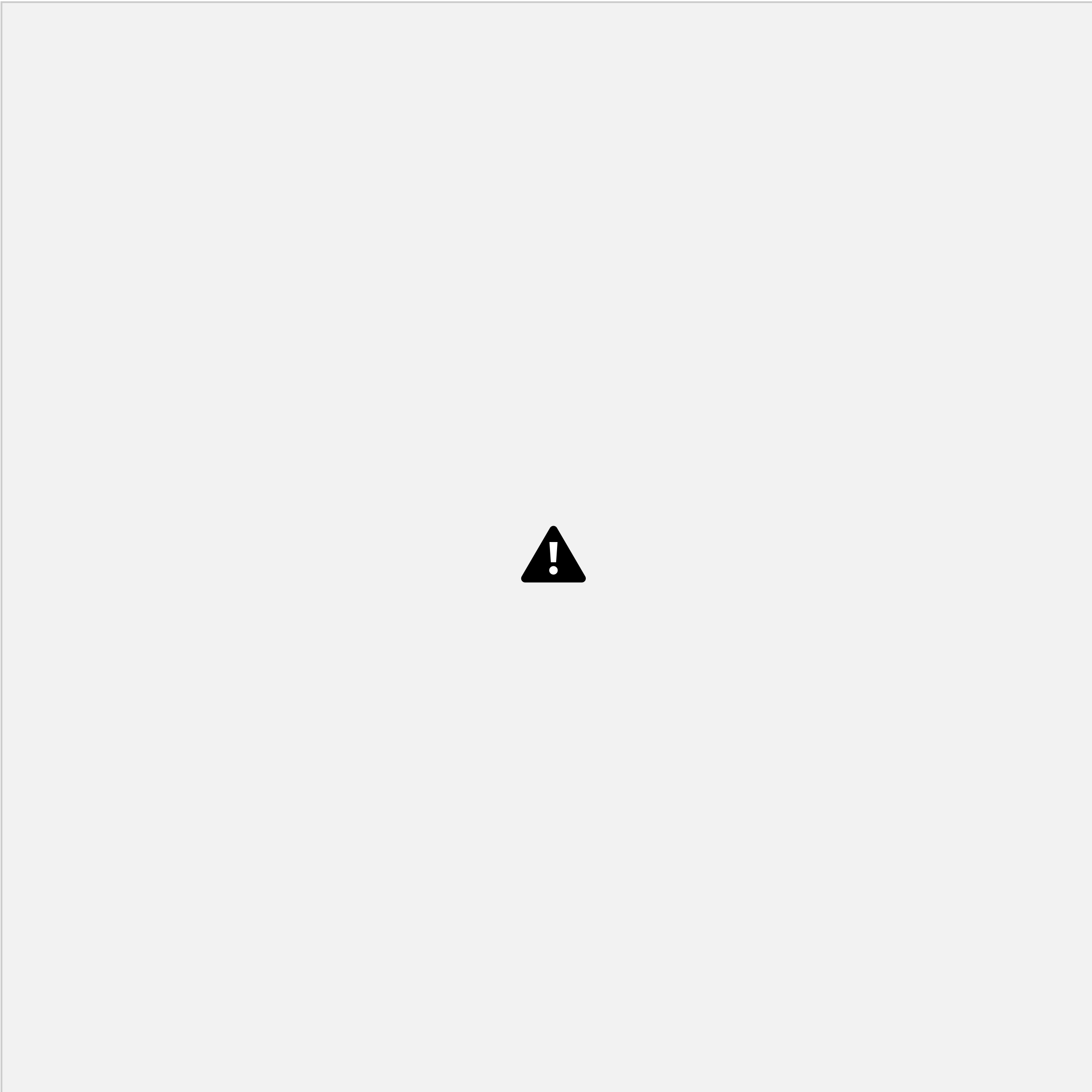


# GROUP-D[EQ]

1. RNA is alkali labile while DNA is alkali resistant.[2012, 2013-S, 2019-S] 2. Denaturation of DNA helps to analyse its structure[2013-S]
3. DNA with higher GC content have relatively higher Tm.[2014] 4.

Presence of deoxy sugar makes DNA more stable.[2014-S]

5.



Synthetic Nucleotides are Used As Drugs.[2015]

6. DNA is more stable than RNA.[2016]

7. Adenine nucleotides have various functions beside making nucleic acids.[2017] 8. Nucleotide analogs are used as anticancer agents.[2018]

9. Ciprofloxacin is an antibiotic but 6-Mercaptopurine is an anticancer drug. [2023]

# NUCLEOTIDE METABOLISM

# GROUP-B

1. Give the metabolic basis of hyperuricemia. How allopurinol useful in lowering the levels of serum uric acid?[7][2011-S]

2. Describe the sources of Nitrogen and carbon atoms of purine ring. Describe the formation of uric acid from purine nucleosides.Enumerate the disorders of purine metabolism.[3+6+3][2015-S]

3. A 55 year old female is complaining of joint pain. Her autoantibody profile is within normal limit. (2+6+2)[2024]

i) Which biochemical investigation should you advice for the patient? Write the physiological reference range of that parameter.ii) Describe the metabolism of that compound. iii) Give a short note on management of this patient.

# GROUP-C[SN]

1. Gout.[2013]
2. Source of nitrogen and carbon atoms of the purine ring.[2013] 3. Purine Salvage Pathway.[2018]
4. Salvage pathway of Purine metabolism with its clinical significance.[2025]

# GROUP-D[EQ]

1. Adenosine deaminase deficiency leads to immunodeficiency state.[2011- S] 2. Synthetic Nucleotides Are Used As Drugs.[2015]
3. Intake of alcohol may aggravate the symptoms of gout.[2016]
4. Allopurinol lowers the uric acid concentration of blood.[2012-S, 2019] 5. Gout is precipitated by alcohol intake.[2012-S, 2016-S, 2018-S, 2021]
6. Renal calculi and formedin hyperparathyroidism.[2025]
7. Blue light phototherapy is effective in neonatal jaundice.[2025]

## *ADVANCE TOPICS*

1.What is Lesch Nyhan Syndrome
2.Regulation and inverters of purine synthesis
3.What is orotic aciduria
4.CPS I vs CPS II



# MOLECULAR BIOLOGY & GENETICS

# GROUP-A

1. Describe the stage of initiation of the translation process with the help of a diagram. State the mechanism of action of the following antibiotics in the inhibition of translation:
A) Streptomycin, B) Erythromycin,
C)Chloramphenicol.[6+6][2011]
2. Write down the different types of DNA damage. Explain the mechanisms of:
Mismatch DNA repair, Base excision repair, Nucleotide excision repair.[6+6][2011]
3. Explain the principles of polymerase chain reaction. Enumerate the reagents and equipments needed for this procedure. What are the applications of PCR in medical sciences?[6+6][2011-S]
4. Describe the enzyme and proteins associated with prokaryotic DNA replication process.[12][2012-S]

5. Define operon. Describe the Lac-operon model for regulation of gene expression in E.Coli.[2+10] [2018]

6. Write with the help of a diagram, describe the stage of initiation and elongation of translation process in E.coli. State the mechanism of streptomycin and puromycin in the inhibition of translation process on prokaryotes.[9+3][2019]

7. Write down with the help of a diagram stage of initiation and elongation of replication process. Differentiate between DNA polymerase I and DNA polymerase III[3+9][2019]

8. Name the different types of DNA damage. Mention different types of DNA repair. With the help of diagram differentiate between ‘Mismatch repair’ & ‘Base Excision Repair’[2+2+8][2019-S]

9. With the help of a suitable diagram explain the ‘Lac-operon model’ of regulation of gene expression in bacteria. Differentiate between the mono cistronic & poly-cistronic mRNA.[10+2][2019-S]

10.Explain replication of a DNA molecule in the context of initiation, elongation and termination in prokaryotes with suitable diagrams. Enumerate the difference b/w DNA polymerase I, II & III.[10+5][2021] 11. A 78year old male was admitted at our hospital with clinical and laboratory features allowing us to make the diagnosis of acute myeloid leukaemia (AML). No evidence of preceding CML (splenomegaly or basophilia) was found. The largotyps an

Ghamist metaphases was 46XY.49,22)(g34:gHt). Molecular analysis suggeses du angical piso c6a2 transcript seems to be associated in AML with aggressive disease i Give a detailed account of the transcription process

How is it regulated?Name inhibitors of transcription.

12.A 46 year old male patient was admitted to the hospital with symptoms of diphthena, condition caused by corynebacterium diphtheria. The diphtheria

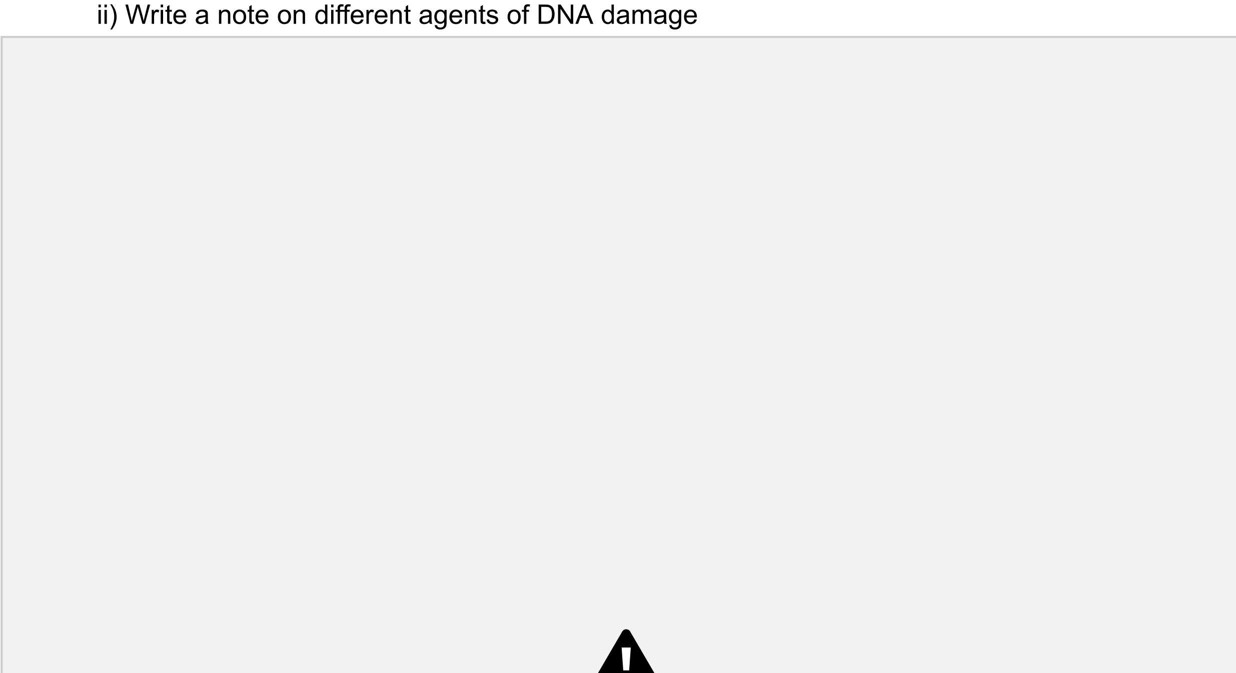


toxin inhibits translation in mammalian systems 1) Describe the process of translation in eukaryotes with flow diagram Name three inhibitors of protein synthesis and mention their mechanism of action. i) Enumerate post translational modifications. [6+3+6] [2022]

13. A 5 year old boy presented with blistering photosensitive lesions diagnosed as xeroderma pigmentosa

i) What is the molecular basis of this disorder?

ii) Write a note on different agents of DNA damage



iii)

Enumerate any six types of DNA repair methods.
iv) Name the prokaryotic DNA polymerases involved in DNA repair
v) Name the eukaryotic DNA polymerases with their roles 2+3+3+3+4 [2023] 14. Outline with diagram the process of elongation phase of translation in prokaryotes. Add a note on inhibitors of translation with examples. Mention three types of post-translational processing. Classify mutation. Explain the consequence of point mutation with a suitable example. 4+3+3+2+3 [2023]

# GROUP-B

1. Using gene transfer technology outline the steps of desired protein synthesis.[7][2010-S] 2. Describe how ribonucleic acid is synthesised. Indicate the difference b/w DNA Polymerase III and RNA polymerase.[5+2][2010]
3. Describe the initiation, elongation and termination phase of transcription in eukaryotes. Name the antibiotics which specifically inhibit the microbial protein synthesis.[5+2][2015 - 12 marks][2013]
4. Give a brief account of different types of mutation with examples.[2014- S] 5. Enumerate different types of DNA repair.[7][2015-S, 2018]
6. Give an account of negative and positive regulation of lac operon in E.Coli.[2015][7] 7. Enumerate the DNA damaging agents and indicate the types of damages made by them.[7][2017]
8. Expand the term PCR. Describe different steps for PCR reaction. Enumerate any four uses of the PCR.[1+7+2][2021]
9. Enumerate the different modes of repair of DNA damage in humans. Explain the mechanism of any one of them with the disorder arising due to the defects on that pathway.[4+4+2][2021]
10. Give an account of the positive and negative regulation of Lac operon in E. coli. [5+5] [2023]
11. What is promoter? Discuss the process of initiation of prokaryotic transcription. Enumerate types of eukaryotic RNA polymerases and mention their individual function.6+4[2024]

# GROUP-C[SN]

1. Restriction Fragment Length Polymorphism (RFLP).[2011]
2. Monoclonal Antibodies.[2011]
3. RNA polymerase[2011-S]
4. Nucleosome[2011-S][2015-S]
5. Polymerase Chain Reaction.[2012-S,2015-S.2018]
6. Base excision repair of DNA.[2013]



7. Frameshift mutation.[2013]
8. Eukaryotic topoisomerase.[2014]
9. Polyclonal Antibodies.[2014]
10. Radioisotopes.[2014]
11. DNA sequencing[2014-S]
12. DNA replication in eukaryotes and prokaryotes.[2015]
13. Post translational modification of protein with examples of their function.[2025] 14. RNA

interference.[2025]

15. Frame shift mutation and its consequences.[2025]

16. RNA editing.[2015] [2023]

17. Use of in vitro DNA amplification process in the laboratory[2015-S] 18.Post translational modification of proteins[2015-S]

19.Point mutation.[2012-S,2016]

17.Restriction endonuclease[2017-S] 18.Eukaryotic DNA polymerases[2017-S]

19.Satellite, minisatellite and microsatellite DNA.[2017-S] 20.Point mutation.[2019]

21.Genome of retrovirus.[2019]

22.Genetic Code.[2019-S]

23.Southern blotting technique[2022]

# GROUP-D[EQ]

1. RNA can act as enzyme.[2011]
2. DNA denaturation is essential for DNA hybridization.[2012-S]
3. Genetic code is degenerate. [2012-S,2018]
4. Replication fork is the main constituents for DNA replication.[2014-S] 5. Sickle Cell anaemia is an example of point mutation[2015-S]
6. DNA is much more stable than RNA.[2016]
7. Ribosome is the ultimate ribozyme.[2017]
8. Recombinant DNA technology is required for selective amplification of a particular gene.[2017-S]
9. RNA editing mechanism is responsible for APO B48 synthesis in the intestinal cells. [2019]
10. Restriction enzymes differ from other DNAases. [2023]
11. Sickle cell anaemia is a molecular disease. [2023]
12. Xeroderma pigmentosa occurs due to defect in DNA repair mechanisms. [2023] 13.Yeast artificial chromosome can act as a high capacity vector in DNA cloning.[2022] 14. Restriction endonuclease show different cleavage patterns [2023] 15. Philadelphia Chromosome in CML is an example of Chromosomal translocation [2023] 16. Topo-isomerase enzyme helps in correcting super-coiling.[2024]
17. PCR is the gold standard for TB germ detection.[2024]
18. Soothern blotting technique is an ideal technique for identifying a DNA segment. [2024]

19. .P53 is khown as Guardian of genome as well as Guardian of cell.[2025]

## ***ADVANCED TOPICS***



1.Draw the structure of trna
2 Types of DNA polymers of eukaryotes and prokaryotes with their function 3. Genetic Code
4. Difference between eukaryotic and prokaryotic translation

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# IMMUNE-CHEMISTRY & CANCER

# GROUP-B

1. Draw the representative structure of IgG. Classify immunoglobin and mention the function of each class.[2017-S][2019] PAGE 745-746

2. Describe the structural characteristics of an Ig molecule in general with a diagram. Explain the functional difference b/w T cell & B cell mediated immunity.[5+5][2021] 3. Briefly describe different types of graft rejection reactions. [2023] PAGE 743-745 4. Name the four different types of hypersensitivity Give ane example of each type of hypersensitivity, Describe the mechanism of types I
hypersensitivity.[2022]

5. Draw the structure of an immunoglobin molecule and mention following regions: 6+4 [2023] i) Amino & carboxy terminal, ii)-S-S-linkages, iii) Fab & Fc segment, iv) Papain & pepsin cleaving sites, v)Variable & constant region, vi) Antigen binding site. Explain briefly: Constant regions determine class specific effector functions of an immunoglobin.

6. A seven year old boy showing small blue round cells consistent with Ewing's sarcoma. i) Mention the best method to confirm translocation on t(11.22).
ii) Mention five different methods of conversion of proto-oncogene to oncogene. iii) Give two examples of tumour suppressor genes. Describe one of them. iv) Name one oncofetal marker.(2+5+2+5+1)[2024]
7. Define cancer. Enumerate the causes of cancer including chemical, physical, biological, genetic and viral carcinogens/mutagens. 2+8[2024]

# GROUP-C[SN]

1. Cell Cycle Regulators.[2010] PAGE 680-682
2. Monoclonal antibody[2010-S] PAGE 753
3. Structure & function of IgG [2011-S][2015-S][2018] PAGE 745,746 4. Tumour markers.[2012-S, 2014,2016-S, 2021] PAGE 779-780
5. Ceruloplasmin.[2014] PAGE 449
6. Proto-oncogenes[2014-S][2018] PAGE 774
7. Oncogenes[2016-S] PAGE 774-775
8. Radioisotope Treatment[2018-S] PAGE 831-832
9. Cell Cycle[2021] PAGE 680-681
10. Immunodeficiency Diseases [2023]
11. Chemical Carcinogens [2023]
12. P53 tumour suppressor gene. [2023]
13. Compare and contrast innate immunity & Acquired Immunity.5.[2024] 14.

Tumor markers and their importance.[2025]

15. Types of Hypersensitivity reaction.[2025]



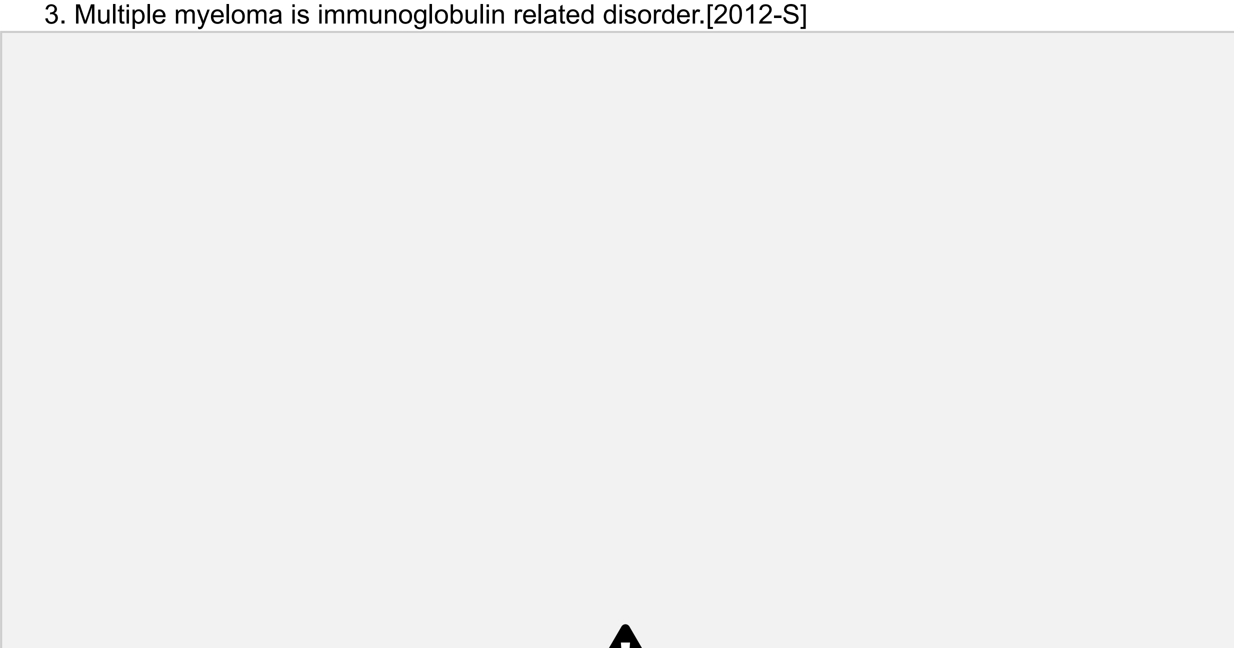
# GROUP-D[EQ]

1. Methotrexate is used for anti-cancer therapy.[2012,2011, 2016-S, 2019- S] 2. Cancer is gene related disease[2011-S]

****Topoisomerase inhibitors work as antibacterial and anticancer agents.[2025].

****Telomerase is involved in aging process and cancer.[2025]

3. Multiple myeloma is immunoglobulin related disorder.[2012-S]



4.

Radioimmunoassay techniques has got demerits also.[2013]
5. Telomeric length is maintained in cancer cells and stem cells.[2016-S] 6. P53 is considered as “Guardian of the genome”.[2019-S]
7. Proto oncogenes are regulatory genes[2021]
8. Under some conditions immunity causes damaging effect[2021]
9. Cancer may be caused by excessive activity of protein tyrosine kinase activity[2022] 10. Apoptosis is very important for preventing cancer. [2023]

# MOLECULAR ENDOCRINOLOGY

## GROUP-A

1. Classify hormones according to their mechanism of action. Indicate the structure and function of thyroid hormone. Explain the inhibitor action of iodine and thiocyanate on thyroid function. Explain mechanism inT3 toxicosis [4+4+2+2][2017- S]
2. List the name of four hormones that act through G-protein coupled receptor complex. Describe the process of signal transduction by any one of those hormones. Briefly state the role of Calcium in signal transduction.[2+8+5][2022]

## GROUP-B

1. In a flow diagram describe how insulin and glucagon regulate the process of lipogenesis and lipolysis in adipose tissue.[7][2010] PAGE 231,232 2. Discuss the different types of G-protein coupled signal transduction processes.[7][2010] PAGE 803
3. Structure and mechanism of action of insulin[7][2012-S] PAGE 187 4. Explain the mechanisms of signal transductions by cAMP, calcium and phosphatidylinositol system with the help of diagrams.[7][2013] PAGE 803-807
5. Chemical structure of insulin & its receptors.[2015-S] PAGE 186-187 6. Classify hormones based on their mechanism of action. Explain second messenger with example, discuss any one in detail. 5+5 [2024]

## GROUP-C[SN]

1. Mode of Action of Steroid Hormone[2010-S] PAGE 806
2. G-Protein[2011] PAGE 803
3. cAMP[2012-S] PAGE 804
4. Insulin Receptor.[2019-S] PAGE 187

## GROUP-D[EQ]

1. Insulin receptor has tyrosine kinase activity. [2010-S]



2. Lipids can act as intracellular signals.[2012]
3. Endocrine disorder may predispose to obesity.[2021]

# EXTRACELLULAR MATRIX

## GROUP-C[SN]

1. Chemical Structure Of Collagen [2010-S,2016] PAGE 786

# GROUP-D[EQ]

1. Ascorbic acid helps in the maturation of the immune collagen molecule.[2019] 2.Post translational modification of collagen confers strength and rigidity [2023]